Muscle Relaxants MCQ – Chapter 13

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 5: General Anesthesia (Chapter 13)

50 Questions • 35 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. Acetylcholine is synthesized in the nerve cytoplasm from choline and acetyl coenzyme A by which enzyme?

- **A.** Acetylcholinesterase
- **B.** Choline acetyltransferase
- **C.** Pseudocholinesterase
- **D.** Monoamine oxidase

Ans: B

Q2. Acetylcholine is destroyed in less than 1 ms at the neuromuscular junction by which enzyme?

- **A.** Pseudocholinesterase (butyrylcholinesterase)
- **B.** Choline acetyltransferase
- **C.** Acetylcholinesterase (true cholinesterase)
- **D.** Catechol-O-methyltransferase

Ans: C

Q3. Which subunit of the acetylcholine receptor is the important site where ACh, succinylcholine and competitive antagonists bind?

- **A.** Beta (β) subunit
- **B.** Alpha (α) subunit
- **C.** Gamma (γ) subunit
- **D.** Delta (δ) subunit

Ans: B

Q4. Which muscle is considered the most resistant to muscle relaxants?

- **A.** Adductor pollicis
- **B.** Diaphragm
- **C.** Orbicularis oculi
- **D.** Laryngeal muscles

Ans: B

Q5. In clinical practice, central muscles are blocked earlier than peripheral muscles (limb muscles). This is because:

- **A.** Central muscles are smaller
- **B.** Of early and more blood supply delivering more relaxant to central muscles
- **C.** Central muscles have fewer receptors
- **D.** Peripheral muscles are more sensitive

Ans: B

Q6. Which muscle is most commonly monitored for neuromuscular blockade?

- **A.** Diaphragm
- **B.** Adductor pollicis (supplied by ulnar nerve)
- **C.** Orbicularis oculi
- **D.** Masseter

Ans: B

Q7. Succinylcholine is metabolized by which enzyme? A. Acetylcholinesterase B. Plasma pseudocholinesterase (butyrylcholinesterase) C. Choline acetyltransferase D. Hoffman degradation	Ans: B
Q8. The excessive repeated depolarization and contractions produced by succinylcholine can be directly visualized as: A. Fasciculations B. Tetany C. Tremor D. Fading	Ans: A
Q9. According to Table 13.1, which feature distinguishes a depolarizing block from a nondepolarizing block? A. It shows fading on neuromuscular monitoring B. It is preceded by muscle fasciculations C. It exhibits post-tetanic facilitation D. It is reversed by neostigmine	Ans: B
Q10. What is the most prominent effect of suxamethonium? A. Bradycardia B. Hyperkalemia C. Increased intraocular tension D. Increased intragastric pressure	Ans: B
Q11. Postoperative myalgia (muscle pains) from succinylcholine can be reduced by precurarization, which involves: A. Giving one-tenth dose of nondepolarizing relaxant 3 minutes prior to succinylcholine B. Giving a double dose of succinylcholine C. Giving neostigmine before succinylcholine D. Hyperventilating the patient	Ans: A
Q12. Succinylcholine is the most commonly implicated drug for which life-threatening condition? A. Anaphylaxis B. Malignant hyperthermia C. Masseter spasm D. Pulmonary edema	Ans: B
Q13. Which serum potassium level is an absolute contraindication for using suxamethonium? A. Serum K+ > 5.5 mEq/L B. Serum K+ > 3.5 mEq/L C. Serum K+ < 3.0 mEq/L D. Serum K+ > 4.0 mEq/L	Ans: A

Q14. Prolonged apnea after succinylcholine can be due to all of the following EXCEPT: **A.** Low pseudocholinesterase **B.** Atypical pseudocholinesterase **C.** Phase II block **D.** Hyperkalemia	**Ans: D**

Q15. Atypical pseudocholinesterase is a genetic disorder diagnosed by which test? **A.** Fluoride number **B.** Dibucaine number **C.** Train of four ratio **D.** Tensilon test	**Ans: B**

Q16. Phase II block (dual block) is produced by an excessive dose of suxamethonium (> 5 mg/kg or > 500 mg). Which finding is pathognomonic of phase II block? **A.** Fasciculations **B.** Succinylcholine showing fading on neuromuscular monitoring **C.** Post-tetanic depression **D.** Hyperkalemia	**Ans: B**

Q17. The onset of action of a nondepolarizer is inversely proportional to its: **A.** Potency **B.** Molecular weight **C.** Duration **D.** Lipid solubility	**Ans: A**

Q18. The basic difference between steroidal and benzylisoquinoline (BZIQ) compounds is that: **A.** Histamine release is seen with BZIQ compounds and vagolytic property with steroidal compounds **B.** Steroidal compounds release histamine **C.** BZIQ compounds are vagolytic **D.** Both groups are identical	**Ans: A**

Q19. The main reason for not preferring pancuronium in current day practice is its: **A.** Cardiac instability (tachycardia and hypertension) **B.** Histamine release **C.** Short duration **D.** Hepatic metabolism	**Ans: A**

Q20. Which nondepolarizing muscle relaxant is the most cardiovascular stable, making it the muscle relaxant of choice for cardiac patients? **A.** Pancuronium **B.** Vecuronium **C.** Atracurium **D.** Gallamine	**Ans: B**

Q21. Which nondepolarizing muscle relaxant is the agent of choice for intubation and precurarization due to its early onset (60–90 seconds)?

A. Vecuronium
B. Pancuronium
C. Rocuronium
D. Atracurium

Ans: C

Q22. Rapacuronium was withdrawn from the market because it produced:

A. Intense bronchospasm in a significant number of patients (> 9%)
B. Severe hyperkalemia
C. Malignant hyperthermia
D. Prolonged apnea

Ans: A

Q23. D-tubocurare is no more used in current practice because of its:

A. Highest propensity for histamine release and ganglion blockade (and hence severe hypotension)
B. Cardiac instability
C. Renal toxicity
D. Short duration of action

Ans: A

Q24. Atracurium undergoes a unique method of degradation which is pH and temperature dependent. What is this called?

A. Ester hydrolysis
B. Hoffman degradation
C. Pseudocholinesterase metabolism
D. Renal elimination

Ans: B

Q25. Because its metabolism is not dependent on hepatic or renal functions, atracurium is the relaxant of choice for:

A. Cardiac patients only
B. Hepatic failure, renal failure, newborn and old age
C. Day care surgery
D. Intubation

Ans: B

Q26. Atracurium's metabolic product laudanosine can cross the blood brain barrier and produce:

A. Convulsions
B. Bronchospasm
C. Hypotension
D. Bradycardia

Ans: A

Q27. The chief advantage of cisatracurium over atracurium is that:

A. It does not release histamine and laudanosine production is 5 times less
B. It has a faster onset
C. It is eliminated by kidneys
D. It is more potent and cardiac stable

Ans: A

Q28. Which nondepolarizing muscle relaxant is metabolized by pseudocholinesterase and is the muscle relaxant of choice for day care surgery? **A.** Mivacurium **B.** Doxacurium **C.** Pancuronium **D.** Pipecuronium	Ans: A

Q29. Gallamine is the only nondepolarizer which can cross the placenta, therefore it is absolutely contraindicated in: **A.** Renal disease **B.** Pregnancy **C.** Hepatic disease **D.** Cardiac disease	Ans: B

Q30. Gantacurium, an onium chlorofumarate under research, is metabolized by forming a cysteine adduct followed by ester hydrolysis. It may emerge as an alternative to which drug? **A.** Atracurium **B.** Suxamethonium for intubation **C.** Vecuronium **D.** Pancuronium	Ans: B

Q31. When two muscle relaxants of the SAME group (steroid + steroid or BZIQ + BZIQ) are added, the effect produced is: **A.** Additive **B.** Synergistic **C.** Antagonistic **D.** No effect	Ans: A

Q32. Inhalational agents prolong the block of muscle relaxants and decrease the requirement of relaxant by: **A.** 20–30% **B.** 50–60% **C.** 5–10% **D.** 70–80%	Ans: A

Q33. Which class of antibiotics prolongs the effect of both depolarizers and NDMR, and is reversed by 4-aminopyridine? **A.** Penicillins **B.** Aminoglycosides **C.** Cephalosporins **D.** Macrolides	Ans: B

Q34. **According to Table 13.2, in conditions with extrajunctional receptors (burns, hemiplegia, paraplegia, denervation), the response to depolarizers and nondepolarizers is:** **A.** Increased sensitivity/hyperkalemia to depolarizers and resistance to nondepolarizers **B.** Resistance to depolarizers and increased sensitivity to nondepolarizers **C.** Increased sensitivity to both **D.** Resistance to both	**Ans: A**

Q35. **In myasthenia gravis, the response to muscle relaxants is:** **A.** Increased sensitivity to depolarizers and resistance to nondepolarizers **B.** Resistance to depolarizers and increased sensitivity to nondepolarizers **C.** Increased sensitivity to both **D.** Resistance to both	**Ans: B**

Q36. **According to the rules of reversal, a reversal agent should be given only after:** **A.** Immediately after the last dose of relaxant **B.** Some evidence of spontaneous recovery has appeared (e.g. spontaneous respiration, TO4 shows at least 1 twitch) **C.** The patient is fully awake **D.** 30 minutes regardless of recovery	**Ans: B**

Q37. **Which anticholinesterase, being NOT a quaternary ammonium compound, can cross the blood brain barrier?** **A.** Neostigmine **B.** Pyridostigmine **C.** Edrophonium **D.** Physostigmine	**Ans: D**

Q38. **Why is glycopyrrolate preferred over atropine when given with cholinesterase inhibitors during reversal?** **A.** Because glycopyrrolate is devoid of central side effects **B.** Because glycopyrrolate is cheaper **C.** Because glycopyrrolate has a faster onset **D.** Because glycopyrrolate increases heart rate more	**Ans: A**

Q39. **Which is the most commonly used reversal agent, and what is its dose?** **A.** Neostigmine, 0.04–0.08 mg/kg **B.** Edrophonium, 0.5 mg/kg **C.** Pyridostigmine, 0.2 mg/kg **D.** Physostigmine, 0.1 mg/kg	**Ans: A**

Q40. **Which is the reversal agent of choice for mivacurium?** **A.** Neostigmine **B.** Pyridostigmine **C.** Edrophonium **D.** Sugammadex	**Ans: C**

Q41. Sugammadex (gamma cyclodextrin) directly binds which type of muscle relaxants to form a complex eliminated unchanged through the kidney? **A.** Benzylisoquinoline type **B.** Steroidal type of nondepolarizing muscle relaxants **C.** Depolarizing type **D.** Centrally acting relaxants	**Ans: B**

Q42. A major advantage of sugammadex over anticholinesterases is that: **A.** It is effective in reversing profound block, while anticholinesterases cannot reverse deeper blocks **B.** It is cheaper **C.** It can reverse benzylisoquinoline relaxants **D.** It can be used in severe renal insufficiency	**Ans: A**

Q43. Which clinical sign is considered the BEST clinical sign of adequate reversal, though difficult to perform, making head lift the most widely used test? **A.** Spontaneous eye opening **B.** Sustained jaw clench over a tongue blade **C.** Recovery of cough reflex **D.** Spontaneous limb movements	**Ans: B**

Q44. A train of four (TO4) ratio of what value indicates adequate recovery, with a ratio above 0.9 guaranteeing recovery? **A.** > 0.7 (70%) **B.** > 0.3 (30%) **C.** > 0.5 (50%) **D.** > 0.1 (10%)	**Ans: A**

Q45. Respiratory acidosis and metabolic alkalosis can double the dose of neostigmine required, and it becomes almost impossible to reverse a patient with pCO2 more than: **A.** 50 mm Hg **B.** 40 mm Hg **C.** 60 mm Hg **D.** 30 mm Hg	**Ans: A**

Q46. Centrally acting muscle relaxants produce muscle relaxation by which mechanism? **A.** By acting at the neuromuscular junction **B.** Through a central mechanism (both supraspinal and spinal level), inhibiting polysynaptic reflexes; they have no action on the NMJ **C.** By blocking acetylcholine receptors **D.** By inhibiting pseudocholinesterase	**Ans: B**

Q47. For the treatment of tetanus, which centrally acting muscle relaxant is most effective? **A.** Intravenous diazepam **B.** Baclofen **C.** Tizanidine **D.** Chlorzoxazone	**Ans: A**

Q48. Suxamethonium increases which pressures in the body? **A.** Only intraocular pressure **B.** Intraocular, intragastric and intracranial pressures **C.** Only intracranial pressure **D.** Only intragastric pressure	**Ans: B**

Q49. Which nondepolarizing muscle relaxant has the longest duration of action? **A.** Vecuronium **B.** Mivacurium **C.** Doxacurium **D.** Rocuronium	**Ans: C**

Q50. Which nondepolarizing muscle relaxants are SAFE in renal failure? **A.** Gallamine, metocurine, doxacurium, pancuronium **B.** Atracurium, cisatracurium, rocuronium, mivacurium **C.** Vecuronium and pancuronium only **D.** Tubocurare and gallamine	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — Choline acetyltransferase

Page 122: Acetylcholine is synthesized in the nerve cytoplasm by combination of choline and acetyl coenzyme A. This reaction is mediated by the enzyme choline acetyltransferase. The synthesized ACh is stored in vesicles.

Topic: Physiology of NMJ

Q2. Answer: C — Acetylcholinesterase (true cholinesterase)

Page 123: The acetylcholine is destroyed in less than 1 ms by acetylcholinesterase (true cholinesterase) which is synthesized by the underlying muscle.

Topic: Physiology of NMJ

Q3. Answer: B — Alpha (α) subunit

Page 123: The alpha (α) subunit is important because ACh, succinylcholine and competitive antagonists (nondepolarizing relaxants) bind to this site. The receptor has two gates — an upper voltage-dependent and a lower time-dependent gate.

Topic: Acetylcholine Receptor

Q4. Answer: B — Diaphragm

Page 123: Laryngeal muscles and diaphragm may require more than double the doses required to block adductor pollicis. In fact, the diaphragm is considered as the most resistant muscle.

Topic: Sequence of Muscle Blockade

Q5. Answer: B — Of early and more blood supply delivering more relaxant to central muscles

Page 123: Central muscles are blocked earlier than peripheral muscles simply because of early and more blood supply delivering more quantity of muscle relaxants to central muscles as compared to peripheral muscles. The sequence of recovery is also the same — central muscles recover first and limb muscles last.

Topic: Sequence of Muscle Blockade

Q6. Answer: B — Adductor pollicis (supplied by ulnar nerve)

Page 123: The most commonly chosen muscle is adductor pollicis, supplied by the ulnar nerve. Relaxation of adductor pollicis means laryngeal muscles have already been blocked and intubation can be performed.

Topic: Neuromuscular Monitoring

Q7. Answer: B — Plasma pseudocholinesterase (butyrylcholinesterase)

Page 124: Succinylcholine is immediately metabolized in plasma by pseudocholinesterase (butyrylcholinesterase), which is synthesized in liver and present in plasma in abundant quantities. To prevent its metabolism, it should be given at a faster rate.

Topic: Suxamethonium

Q8. Answer: A — Fasciculations

Page 124: Excessive succinylcholine produces repeated depolarization and contractions which can be directly visualized as fasciculations. Persistent depolarization makes the membrane refractory, producing what is termed phase I block.

Topic: Suxamethonium

Q9. Answer: B — It is preceded by muscle fasciculations

Page 124, Table 13.1: A depolarizing block (phase I block) is preceded by muscle fasciculations, shows no fading, no post-tetanic facilitation, and is not reversed by cholinesterase inhibitors. A nondepolarizing block shows no fasciculations, fading, post-tetanic facilitation, and is reversed by neostigmine.

Topic: Suxamethonium

Q10. Answer: B — Hyperkalemia

Page 125: Hyperkalemia is the most prominent effect of suxamethonium, which occurs due to excessive muscle fasciculations. In normal circumstances serum potassium increases by 0.5 mEq/L. Hyperkalemia increases the risk of ventricular arrhythmias.

Topic: Suxamethonium

Q11. Answer: A — Giving one-tenth dose of nondepolarizing relaxant 3 minutes prior to succinylcholine

Page 125: Precurarization is the technique in which one-tenth dose of nondepolarizing muscle relaxant is given 3 minutes prior to succinylcholine. This small dose decreases fasciculations and hence myalgia. Incidence of muscle pains is 40–50%.

Topic: Suxamethonium

Q12. Answer: B — Malignant hyperthermia

Page 125: Malignant hyperthermia — succinylcholine is the most commonly implicated drug. Masseter spasm with succinylcholine, especially in children, may be a precursor sign for malignant hyperthermia.

Topic: Suxamethonium

Q13. Answer: A — Serum K+ > 5.5 mEq/L

Page 125: Hyperkalemia — serum K+ > 5.5 mEq/L is an absolute contraindication for using suxamethonium, because suxamethonium further increases serum potassium.

Topic: Suxamethonium

Q14. Answer: D — Hyperkalemia

Page 126: Prolonged apnea after succinylcholine can be because of low pseudocholinesterase, atypical pseudocholinesterase, or phase II block.

Topic: Suxamethonium

Q15. Answer: B — Dibucaine number

Page 126: Atypical (abnormal) pseudocholinesterase is a genetic disease in which the patient has abnormal enzyme not able to metabolize suxamethonium. Incidence is 1 in 3,000. This is diagnosed by the dibucaine number. Dibucaine inhibits 80% of normal enzyme and 20% of abnormal enzyme.

Topic: Suxamethonium

Q16. Answer: B — Succinylcholine showing fading on neuromuscular monitoring

Page 127: Phase II block (also called dual block) is prolonged block seen after excessive dose of suxamethonium (> 5 mg/kg or > 500 mg). Succinylcholine showing fading on neuromuscular monitoring is pathognomonic of phase II block.

Topic: Suxamethonium

Q17. Answer: A — Potency

Page 127–128: Onset of a nondepolarizer is inversely proportional to potency. More the potency, lesser the dose required, means less molecules delivered at the neuromuscular junction to compete with acetylcholine, delaying the onset of action.

Topic: Nondepolarizing Muscle Relaxants

Q18. Answer: A — Histamine release is seen with BZIQ compounds and vagolytic property with steroidal compounds

Page 127: The basic difference between steroidal and benzylisoquinoline (BZIQ) compounds is that histamine release is seen with BZIQ compounds and vagolytic property with steroidal compounds.

Topic: Nondepolarizing Muscle Relaxants

Q19. Answer: A — Cardiac instability (tachycardia and hypertension)

Page 128: The main reason for not preferring pancuronium is its cardiac instability. It releases nor-adrenaline and causes vagal blockade, leading to tachycardia and hypertension. By releasing nor-adrenaline it increases the chances of arrhythmias with halothane.

Topic: Steroidal Compounds

Q20. Answer: B — Vecuronium

Page 128: As vecuronium is the most cardiovascular stable, it is the muscle relaxant of choice for cardiac patients. Long-term use in ICU has resulted in polyneuropathy due to accumulation of its metabolite, 3-hydroxy metabolite.

Topic: Steroidal Compounds

Q21. Answer: C — Rocuronium

Page 128: The advantage of rocuronium over other nondepolarizing muscle relaxants is its early onset (within 60–90 seconds compared to 3–4 minutes for other relaxants). As its onset is comparable to succinylcholine, it is the nondepolarizer of choice for intubation and precurarization. It is also the only nondepolarizing muscle relaxant which can be given by intramuscular route.

Topic: Steroidal Compounds

Q22. Answer: A — Intense bronchospasm in a significant number of patients (> 9%)

Page 128: Rapacuronium is the analog of vecuronium with less potency. It has been withdrawn from the market because it produced intense bronchospasm in a significant number of patients (> 9%).

Topic: Steroidal Compounds

Q23. Answer: A — Highest propensity for histamine release and ganglion blockade (and hence severe hypotension)

Page 128: D-tubocurare is not used in present day practice because of its highest propensity for histamine release and ganglion blockade (and hence severe hypotension). It was named so because it was carried in bamboo tubes and used as arrow poison by Amazon people.

Topic: Benzylisoquinoline Compounds

Q24. Answer: B — Hoffman degradation

Page 129: The majority of atracurium undergoes spontaneous degradation in plasma called Hoffman degradation. Hoffman degradation is a pH and temperature-dependent reaction; higher pH and temperature favor more degradation.

Topic: Benzylisoquinoline Compounds

Q25. Answer: B — Hepatic failure, renal failure, newborn and old age

Page 129: As atracurium metabolism is not dependent on hepatic or renal functions, it is the relaxant of choice for hepatic failure, renal failure, newborn (immature hepatic/renal function), old age, and when reversal agent is contraindicated.

Topic: Benzylisoquinoline Compounds

Q26. Answer: A — Convulsions

Page 129: At higher doses, atracurium's metabolic product laudanosine can cross the blood brain barrier and can produce convulsions. As it is a benzylisoquinoline compound, it can release histamine producing allergic reactions ranging from pruritic rash to angioneurotic edema, bronchospasm or hypotension.

Topic: Benzylisoquinoline Compounds

Q27. Answer: A — It does not release histamine and laudanosine production is 5 times less

Page 129: Cisatracurium is an isomer of atracurium with 4 times more potency. The chief advantage of cisatracurium over atracurium is that it does not release histamine and laudanosine production is 5 times less than atracurium, therefore always preferred over atracurium. It is only metabolized by Hoffman degradation.

Topic: Benzylisoquinoline Compounds

Q28. Answer: A — Mivacurium

Page 129: Like succinylcholine, mivacurium is metabolized in plasma by pseudocholinesterase, therefore prolonged block is expected in conditions causing low pseudocholinesterase. Its short duration of action (5–10 minutes) makes it a muscle relaxant of choice for day care surgery. The major side effect is histamine release.

Topic: Benzylisoquinoline Compounds

Q29. Answer: B — Pregnancy

Page 129: Gallamine is the only nondepolarizer which can cross the placenta and cause fetal death, therefore absolutely contraindicated in pregnancy. More than 80% is excreted by kidneys, therefore it is also contraindicated in renal disease. It has maximum propensity for vagal blockade (vagolytic), causing severe tachycardia.

Topic: Benzylisoquinoline Compounds

Q30. Answer: B — Suxamethonium for intubation

Page 130: Gantacurium (previously AV430A) is a mixed onium chlorofumarate having fast onset (< 90 seconds) and ultra short duration (shortest among nondepolarizers). It may emerge as an alternative to suxamethonium for intubation. It is rapidly metabolized by forming a cysteine adduct followed by ester hydrolysis.

Topic: Compounds Under Research

Q31. Answer: A — Additive

Page 130: Two muscle relaxants of the same group (steroid + steroid or BZIQ + BZIQ) produce an additive effect, while addition of different groups (steroid + BZIQ) produces a synergistic effect.

Topic: Drug Interactions

Q32. Answer: A — 20–30%

Page 130: Inhalational agents prolong the block by both depolarizers and NDMR and decrease the requirement of relaxant by 20–30%. Maximum relaxation among inhalational agents is produced by desflurane (Desflurane > Sevoflurane > Isoflurane > Halothane).

Topic: Drug Interactions

Q33. Answer: B — Aminoglycosides

Page 130: Antibiotics prolong the effect of both depolarizers and NDMR. Aminoglycosides (neomycin, streptomycin, gentamicin, kanamycin, polymyxin, tobramycin), lincomycin and clindamycin are implicated. To reverse the effect of block by aminoglycosides, the drug used is 4-aminopyridine.

Topic: Drug Interactions

Q34. Answer: A — Increased sensitivity/hyperkalemia to depolarizers and resistance to nondepolarizers

Page 131, Table 13.2: In conditions with extrajunctional receptors (burns, hemiplegia, paraplegia, denervation muscle injuries, tetanus, cerebral palsy), there is increased sensitivity and hyperkalemia to depolarizers but resistance to nondepolarizers, because extrajunctional receptors are sensitive to depolarizing agents and resistant to nondepolarizers.

Topic: Muscular Diseases

Q35. Answer: B — Resistance to depolarizers and increased sensitivity to nondepolarizers

Page 131, Table 13.2: In myasthenia gravis there is resistance to depolarizers and increased sensitivity to nondepolarizers. This autoimmune disease destroys acetylcholine receptors, leaving fewer receptors for succinylcholine (increasing resistance) but less competition for nondepolarizers (increasing their sensitivity).

Topic: Muscular Diseases

Q36. Answer: B — Some evidence of spontaneous recovery has appeared (e.g. spontaneous respiration, TO4 shows at least 1 twitch)

Page 131: Reversal should be given only after some evidence of spontaneous recovery has appeared, like the patient has started spontaneous respiration and TO4 shows at least 1 twitch; otherwise reversal agents may enter the open channels and themselves produce block.

Topic: Reversal of Block

Q37. Answer: D — Physostigmine

Page 131: Anticholinesterases used for reversal are neostigmine, pyridostigmine, edrophonium and physostigmine. All except physostigmine are quaternary ammonium compounds and therefore do not cross the blood brain barrier (physostigmine does).

Topic: Reversal of Block

Q38. Answer: A — Because glycopyrrolate is devoid of central side effects

Page 132: To prevent muscarinic side effects (bradycardia, bronchospasm, secretions), atropine or glycopyrrolate has to be given with cholinesterase inhibitors. Glycopyrrolate, being devoid of central side effects, is always preferred over atropine.

Topic: Reversal of Block

Q39. Answer: A — Neostigmine, 0.04–0.08 mg/kg

Page 132: Neostigmine is the most commonly used reversal agent. Dose is 0.04–0.08 mg/kg. The effect begins in 5–10 minutes and peaks in half hour and lasts for one hour. The anticholinergic preferred with neostigmine is glycopyrrolate.

Topic: Reversal of Block

Q40. Answer: C — Edrophonium

Page 132: As neostigmine and pyridostigmine inhibit pseudocholinesterase, they cannot be used to reverse mivacurium (which is metabolized by pseudocholinesterase). Edrophonium does not inhibit pseudocholinesterase, therefore it is the reversal agent of choice for mivacurium.

Topic: Reversal of Block

Q41. Answer: B — Steroidal type of nondepolarizing muscle relaxants

Page 132: Sugammadex directly binds to the steroidal type of nondepolarizing muscle relaxants to form a complex which gets eliminated unchanged through the kidney. Its major disadvantage is that it cannot reverse benzylisoquinoline type of nondepolarizers.

Topic: Reversal of Block

Q42. Answer: A — It is effective in reversing profound block, while anticholinesterases cannot reverse deeper blocks

Page 132: Anticholinesterases are unable to reverse deeper blocks, while sugammadex is effective in reversing profound blockade. The reversal of block with cholinesterase inhibitors is slow (10–15 min) while reversal with sugammadex is very fast (2–3 minutes). It also has no muscarinic side effects, so there is no need to use glycopyrrolate or atropine.

Topic: Reversal of Block

Q43. Answer: B — Sustained jaw clench over a tongue blade

Page 132: Sustained jaw clench over a tongue blade is considered the best clinical sign of adequate reversal; however, it is difficult to perform by the patient, therefore head lift (ability to lift head for more than 5 seconds) is still the most widely used clinical test.

Topic: Signs of Adequate Reversal

Q44. Answer: A — > 0.7 (70%)

Page 132: Train of four (TO4) ratio > 0.7 (70%) indicates adequate recovery, but a ratio > 0.9 guarantees recovery. Most clinicians believe that in addition to TO4 ratio > 0.9, patients must not exhibit any clinical symptom/sign of impaired neuromuscular recovery. General weakness is considered as the most sensitive symptom of inadequate reversal.

Topic: Signs of Adequate Reversal

Q45. Answer: A — 50 mm Hg

Page 133: Acid base abnormalities — respiratory acidosis and metabolic alkalosis double the dose of neostigmine. It becomes almost impossible to reverse a patient with pCO_2 more than 50 mm Hg.

Topic: Common Causes of Inadequate Reversal

Q46. Answer: B — Through a central mechanism (both supraspinal and spinal level), inhibiting polysynaptic reflexes; they have no action on the NMJ

Page 133: Centrally acting muscle relaxants produce muscle relaxation through a central mechanism both at supraspinal and spinal level. Polysynaptic reflexes involved in maintenance of muscle tone are inhibited at both spinal and supraspinal level. These drugs have no action on the neuromuscular junction.

Topic: Centrally Acting Muscle Relaxants

Q47. Answer: A — Intravenous diazepam

Page 133: Among centrally acting muscle relaxants, for tetanus, intravenous diazepam is most effective. They are also used for muscle spasms, spastic neurological diseases (cerebral palsy, spinal injuries), and closed reductions/dislocations in orthopedics.

Topic: Centrally Acting Muscle Relaxants

Q48. Answer: B — Intraocular, intragastric and intracranial pressures

Page 125: Intraocular, intragastric and intracranial pressures are all increased by suxamethonium. Ocular muscles undergo tonic contraction increasing intraocular tension; neck muscle contraction blocks jugular venous outflow increasing intracranial tension; and contraction of abdominal muscles increases intragastric pressure.

Topic: Suxamethonium

Q49. Answer: C — Doxacurium

Page 128–129: Doxacurium has a long duration of action (60 minutes), the longest acting nondepolarizer, with high potency and almost complete elimination as unchanged drug by kidneys, making it a least preferred drug by clinicians.

Topic: Nondepolarizing Muscle Relaxants

Q50. Answer: B — Atracurium, cisatracurium, rocuronium, mivacurium

Page 134: NDMR safe in renal failure are atracurium, cisatracurium, rocuronium and mivacurium. NDMR absolutely contraindicated in renal failure are gallamine, metocurine, doxacurium, pancuronium and tubocurare.

Topic: Nondepolarizing Muscle Relaxants